FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 3.4: ELECTROSTATICS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION ASSESSMENT — STUDENT TASK

Driving Question: How do invisible electric charges build up, exert forces across empty space, and store energy — and how can we harness these abilities to protect communities and power technology in Kenya?

YOUR TASK: Write your Final Explanation to answer the driving question completely. Use the five sections below as your guide. For each section, read the prompt carefully, study the exemplar response, and then write your own answer in your own words.

A strong Final Explanation does ALL of the following:

- Refers back to the anchoring phenomenon — the lightning strike at Ol Kalou Market in Nyandarua County.
- Uses correct Physics vocabulary in every section.
- Connects evidence from lessons, practicals, and observations to your reasoning.
- Explains not just WHAT happened but WHY and HOW — using atomic-level thinking.
- Shows how electrostatics concepts apply to real Kenyan contexts (lightning conductors, technology, community safety).

Before you write, ask yourself: "If a trader at Ol Kalou Market asked me to explain everything that happened during that storm, could I answer clearly and completely?" Your Final Explanation should be your best, most complete answer to that question.

Section 1 — Introduction: What Puzzled Me and What I Now Understand	
Describe the anchoring phenomenon — the lightning strike at Ol Kalou Market — and explain what puzzled you at the start of the unit. What questions	At the beginning of this unit, I watched a video reconstruction of a lightning strike at Ol Kalou Market in Nyandarua County. A bolt of lightning hit a metal-roofed market stall, leaving a scorch mark and melting a section of iron sheet. The wooden stall right next to it was completely untouched. Traders described the hairs on their arms standing on end just before the strike, and a nearby transistor radio crackled with static. Thirty metres away, a pointed metal lightning conductor on the district offices directed the discharge safely into the ground — no damage at all.

could you NOT answer then? Then summarise, in your own words, what you now understand after nine lessons. Do not fully explain your ideas yet — save the details for Sections 2–5. This section sets the stage for everything that follows.	At first, I was full of questions I could not answer. What invisible 'thing' built up in the storm cloud and in the iron roof? Why did the wooden stall survive when it was so close? How did the lightning conductor 'know' where to send the charge? Why did the radio crackle before any visible lightning appeared? After nine lessons of investigations, practicals, and discussions, I can now answer all of these questions. I understand that electric charges originate from the transfer of electrons between atoms, that materials behave very differently depending on whether electrons can move freely through them, and that charged objects exert forces on each other — and even on neutral objects — across empty space. I understand how charge builds up, how it distributes on different shapes of conductors, how it can be detected using a leaf electroscope, and how the electric force follows precise mathematical rules described by Coulomb's Law. Most importantly, I can explain how all of these ideas connect to protecting the traders at Ol Kalou Market and to powering technology used across Kenya.

Section 2 — The Origin and Nature of Electric Charge	
Explain where electric charges come from, using your knowledge of atomic structure. In your answer: (a) describe the structure of an atom and identify which particles carry charge; (b) explain what happens at the atomic level when two materials are rubbed together; (c) define the Law of Conservation of Charge and give one example from your lessons; and (d) connect your explanation back to what happened in the storm cloud above Ol Kalou — how did charge build up there?	Electric charges originate at the atomic level. Every atom contains a nucleus made of protons, which carry a positive charge, and neutrons, which carry no charge. Surrounding the nucleus are electrons, which carry a negative charge. In a neutral atom, the number of protons equals the number of electrons, so the atom has no overall charge. When two different materials are rubbed together — for example, when a polythene rod is rubbed with a woollen cloth — electrons are transferred from one material to the other. The material that gains electrons ends up with more electrons than protons, so it becomes negatively charged. The material that loses electrons ends up with fewer electrons than protons, so it becomes positively charged. It is crucial to understand that we are not creating new charges — we are only moving electrons that already existed. This is the Law of Conservation of Charge: the total amount of electric charge in a closed system always stays the same. In our practical, when the polythene rod became −2 units of charge, the woollen cloth became +2 units of charge — the total remained zero, just as it was before rubbing began. Above Ol Kalou Market during the storm, a similar process happened on a massive scale. As ice crystals and water droplets in the storm cloud collided violently in strong updrafts, electrons were transferred between them. Lighter, negatively charged ice particles were carried to the top of the cloud; heavier, positively charged particles sank to the bottom. Over minutes, an enormous separation of charge built up — millions and millions of extra electrons concentrated at the base of the cloud. This is the invisible 'thing' that the traders could not see but could feel — the massive electric charge that was about to discharge as lightning.

Section 3 — Conductors, Insulators, and Charge Distribution	
Explain how the behaviour of electric charges depends on the material they are in. In your answer: (a) define conductors and insulators in terms of electron mobility and give two Kenyan examples of each; (b)	Materials can be classified by how easily electrons move through them. In conductors, the outermost electrons of each atom are loosely held and can move freely from atom to atom throughout the material. Metals are the best conductors — the iron sheet roofing at Ol Kalou Market and the copper wiring inside a Nairobi household are both examples. In insulators, electrons are tightly bound to their atoms and cannot move freely. Wood, rubber, glass, and dry sisal rope are all insulators. This difference in electron mobility is what separates a conductor from an insulator. When charge is placed on a conductor, the free electrons repel one another and spread as far apart as possible. On a smooth,

explain why charge distributes evenly on a smooth spherical conductor but concentrates at sharp points; (c) use these ideas to explain why the metal roof at Ol Kalou was struck while the wooden stall next to it was untouched; and (d) explain how charge distribution on a pointed conductor is the key to how a lightning conductor works.	spherical conductor — like the dome of a Van de Graaff generator — this means the charge distributes evenly across the entire outer surface. However, on a conductor with a sharp point or edge, the geometry forces charges to crowd together at that point. The concentration of charge at a sharp tip is much higher than anywhere else on the surface, and this creates an extremely strong electric field at the tip. At Ol Kalou Market, the iron sheet roof was a conductor. When the negatively charged base of the storm cloud passed overhead, free electrons in the iron roof were repelled downward and positive charge was left on the upper surface of the roof, closest to the cloud. This process — called electrostatic induction — meant the roof became the nearest point of opposite charge to the cloud, making it the target for the lightning discharge. The wooden stall next door is an insulator — its electrons could not move freely in response to the cloud's charge, so no separation of charge occurred, and it presented no attractive target for the lightning bolt. A lightning conductor is a thick copper rod with a very sharp pointed tip fixed to the highest point of a building, connected by a thick copper cable to a metal plate buried deep in the earth. Because the tip is so sharp, any charge induced on the conductor by an approaching storm cloud concentrates intensely at the point. This creates a continuous, controlled pathway of ionised air — called a corona discharge — that bleeds charge safely between the cloud and the ground before a destructive lightning bolt can form. On the Nyandarua district offices, this is exactly what happened: the pointed conductor collected the charge and routed it harmlessly into the earth, protecting the building entirely.

Section 4 — Electric Force and Coulomb's Law	
Explain how charged objects exert forces on one another and on neutral objects, even across empty space. In your answer: (a) state the two rules of electrostatic force (like charges and unlike charges); (b) state Coulomb's Law in words and in symbols, defining every variable and its SI unit; (c) explain what the inverse-square relationship means in plain language — what happens to the force if you double the distance? (d) use Coulomb's Law to explain a specific observation from your lessons or from Ol Kalou (e.g., why the traders' arm hairs stood on end before the strike, or how the leaf electroscope responded to a charged rod brought near without touching).	One of the most remarkable things about electric charge is that charged objects can push or pull other objects without touching them — the force acts across empty space. The rules governing the direction of this force are: like charges repel each other (two negative charges push apart; two positive charges push apart), and unlike charges attract each other (a positive charge and a negative charge pull toward one another). The precise mathematical relationship is described by Coulomb's Law, named after French physicist Charles-Augustin de Coulomb. The law states: the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges, and inversely proportional to the square of the distance between them. In symbols: $F = k \times (Q_1 \times Q_2) / r^2$ Where: F is the electrostatic force in Newtons (N); Q_1 and Q_2 are the magnitudes of the two charges in Coulombs (C); r is the distance between the centres of the two charges in metres (m); and k is Coulomb's constant, equal to 9×10^9 N·m²/C². The inverse-square relationship is very important. It means that if you double the distance between two charges, the force between them does not simply halve — it falls to one quarter of its original value (because $2^2 = 4$). If you triple the distance, the force falls to one ninth. The force drops off rapidly as charges move apart. We can use this to explain two observations from Ol Kalou. First, as the negatively charged base of the storm cloud moved closer and closer to the iron roof, the distance r decreased rapidly. By Coulomb's Law, the attractive force between the cloud's negative charge and the induced positive charge on the roof grew enormously — eventually becoming strong enough to ionise the air and drive a lightning bolt across the gap. Second, consider the leaf electroscope in our classroom. When a negatively charged ebonite

	rod was brought near — but not touching — the metal cap, the free electrons in the cap were repelled downward into the leaves. Both leaves received excess negative charge. Since like charges repel, the leaves diverged (spread apart). The closer the rod was brought, the stronger the force (Coulomb's Law), and the wider the leaves spread. This is exactly the same physics — just on a much smaller scale than the Ol Kalou strike.

Section 5 — Bringing It All Together: Answering the Driving Question	
Now bring everything together in one coherent, complete explanation. Answer the full driving question: How do invisible electric charges build up, exert forces across empty space, and store energy — and how can we harness these abilities to protect communities and power technology in Kenya? Your answer must: (a) trace the complete story of the Ol Kalou lightning strike from electron transfer in the cloud all the way to the safe discharge through the lightning conductor; (b) explain at least one other application of electrostatics that is relevant to Kenya (e.g., electrostatic precipitators reducing air pollution from Nairobi's industries, photocopiers, or medical devices); and (c) reflect briefly on how your thinking changed from the start of the unit to now — what was the most surprising or important idea you learned?	I can now tell the complete story of what happened at Ol Kalou Market — from the invisible world of electrons all the way to the scorch mark on the iron roof. It began high in the storm cloud above Nyandarua County. Violently colliding ice particles transferred electrons between them, creating a massive separation of charge: an enormous excess of negative charge at the base of the cloud, and positive charge at the top. This charge did not 'appear from nowhere' — it was there all along in the atoms of water and ice. The Law of Conservation of Charge was never broken; electrons simply moved. As the negatively charged cloud base drifted over Ol Kalou Market, it exerted an electrostatic force — described perfectly by Coulomb's Law — on the free electrons in the metal iron-sheet roof below. Those electrons were pushed away from the cloud, leaving a layer of positive charge on the upper surface of the roof. The wooden stall next door could not respond this way because wood is an insulator; its electrons were bound in place, so no charge separation occurred and it was never a target. Meanwhile, 30 metres away, the pointed copper lightning conductor on the district offices was also responding. Its sharp tip concentrated the induced charge intensely, creating a powerful electric field that ionised the surrounding air and formed a safe, continuous discharge channel. When the potential difference between cloud and ground finally overcame the resistance of the air, the massive discharge — lightning — followed the path of least resistance: down the lightning conductor and into the earth. The iron roof, lacking this controlled pathway, received the full destructive strike instead. The traders' arm hairs stood on end because their bodies — standing on the ground — also experienced electrostatic induction. The repelled electrons moved toward their feet, and the induced positive charge on the surface of their skin and hair was attracted toward the negative cloud above. Individual hairs, each carrying the same induced charge, repelled one another and stood erect. The transistor radio crackled because the rapidly changing electric field around the building induced oscillating currents in its antenna — electrical noise that the speaker reproduced as static. Beyond lightning protection, electrostatics has many important applications for Kenya. Electrostatic precipitators — fitted to the chimneys of industrial plants and cement factories, such as those operating in the Rift Valley — use charged plates to attract and remove dust and ash particles from exhaust gases before they enter the atmosphere. This protects the health of communities living near these facilities. Photocopiers and laser printers, used in every school and government office from Mombasa to Kisumu, use a carefully controlled pattern of static charge on a drum to pick up ink and transfer it precisely onto paper. These technologies all depend on the same principles I studied in this unit. At the start of this unit, I thought electricity was only about wires, switches, and batteries — something that needed to flow in a circuit to exist. The most surprising and important idea I learned is that electric charge can exist, exert enormous forces, and do significant work without any circuit at all. The invisible build-up of electrons in a storm cloud contains enough energy to melt iron — and a simple pointed copper rod, grounded correctly, is enough to harness that energy and protect an entire community. Physics

	that begins with electrons in atoms ends with people in Ol Kalou Market going home safely.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Anchoring Phenomenon Connection	Every section explicitly and accurately connects back to the Ol Kalou lightning strike phenomenon. The student traces a coherent, complete story from electron transfer in the storm cloud to the safe discharge through the lightning conductor, referencing specific details (iron roof, wooden stall, traders' arm hairs, radio static, lightning conductor on district offices).	Most sections reference the Ol Kalou phenomenon clearly and accurately. The student explains the lightning strike story with reasonable completeness but may omit one or two specific details (e.g., does not explain the radio static or the arm hairs).	The student mentions the Ol Kalou phenomenon only in the introduction or conclusion. Connections between lesson content and the phenomenon are vague, superficial, or missing from one or more core sections.
Accuracy and Depth of Scientific Concepts	All five concepts are explained accurately and in depth: atomic structure and electron transfer; Law of Conservation of Charge; conductors vs. insulators (electron mobility); charge distribution and concentration at sharp points; Coulomb's Law (stated in words AND symbols, all variables defined with correct SI units, inverse-square relationship explained clearly). No misconceptions are present.	At least four of the five concepts are explained accurately. Coulomb's Law is stated correctly in symbols but one variable may be missing a unit, or the inverse-square relationship is described but not applied to a specific example. Minor misconceptions are present but do not undermine the overall explanation.	Fewer than three concepts are explained accurately, OR significant misconceptions are present (e.g., stating that rubbing 'creates' charge, or confusing charge sign with charge amount). Coulomb's Law may be missing or incorrectly stated.
Use of Evidence from Lessons and Practicals	The student explicitly cites specific evidence from at least four different lessons or practicals (e.g., polythene rod and woollen cloth practical, leaf electroscope observation, Van de Graaff generator demonstration, Coulomb's Law calculation exercise) and uses that evidence to support each claim. Evidence is clearly linked to reasoning.	The student cites evidence from at least two or three lessons or practicals. Evidence is linked to reasoning in most sections but one or two claims are made without cited support.	The student makes claims with little or no reference to specific lesson evidence. Statements read as memorised facts rather than conclusions drawn from observations and practicals.
Application to Kenyan and Real-World Contexts	The student applies electrostatics concepts accurately to at least two real-world Kenyan contexts — the Ol Kalou lightning conductor is treated as a full case study, AND at least one additional application is described accurately (e.g., electrostatic precipitators in Rift Valley industries, photocopiers in Kenyan	The student applies concepts accurately to the Ol Kalou lightning conductor and mentions one additional application, but the additional application is described at a surface level without a full scientific explanation.	The student discusses the lightning conductor but does not fully explain it using electrostatics principles, OR no additional Kenyan or real-world application is included.

	offices, or medical devices). Explanations show genuine understanding, not just naming of applications.		
Scientific Communication and Vocabulary	Throughout all five sections, the student uses precise Physics vocabulary correctly and consistently: electron transfer, electrostatic induction, conductor, insulator, electron mobility, Coulomb's Law, inverse-square law, electrostatic force, charge distribution, corona discharge, potential difference, Law of Conservation of Charge. Writing is clear, logically structured, and flows from one section to the next as a coherent argument. The driving question is answered completely and explicitly in Section 5.	Most key vocabulary terms are used correctly. A few terms may be used imprecisely or interchangeably (e.g., 'electricity' used instead of 'charge'). The explanation is generally well-structured but the logical flow between some sections is unclear. The driving question is substantially answered.	Key vocabulary is frequently missing, used incorrectly, or replaced with everyday language (e.g., 'the electricity jumped across'). The structure is difficult to follow and the driving question is only partially answered or not addressed directly.